

with SARA-R412M module and cell-phone is used to visualise the analytics platform.

FlashNET IntelliLIGHT is a backend connectivity platform with a Web-based remote monitoring graphic user interface (GUI). Wireless connectivity modules enable a plug-and-play user experience for smart city applications with the following security features:

1. 3GPP global standards compliant (end-to-end connectivity and full two-way manageable channels)

2. Security based on end-to-end 3GPP encryption and guaranteed quality of service (immune to interference)

3. Device interoperability enabled due to use of open standard protocols

4. Scalability and fast development with multiple horizontal IoT platforms

Smartlights can be used at bus stands for safety and surveillance. Wi-Fi provides Internet connectivity to devices by creating a Wi-Fi zone in that area. Cameras keep track of the arrival and departure of buses. The smartlights provide a 24x7 surveillance feed, which can prove beneficial in case of emergencies or criminal activities.

Two-way call buttons on smartlights allow connectivity to the control room or the police during emergency situations. These can also be used by people to talk to someone in the control room for any queries.

The public announcement system in smart streetlights provides a ready way to address a large number of people at a given location. It can be used to announce delays in bus timings and routes, travel distances and more, so that people can make informed decisions.

Wireless technologies to control smart lights

Motion/radar sensors can detect motion in a room, while ambient light sensors can be used to adapt the light to current conditions. Smartphones and tablets can be connected to the mesh network, enabling wireless control of lights through an app. Lights can be connected using short-range radio mesh technologies such as Bluetooth Mesh, Thread, ZigBee or other proprietary technologies like Wirepas Mesh.

Sensor data can be used as input for artificial intelligence (AI) for advanced lighting control and other smart build-



Fig. 2: Smart indoor lighting (Credit: <https://smartiots.wordpress.com>)

ing applications. Short-range radio mesh network can be connected to the cloud and back-end lighting control services through a cellular gateway that supports technologies like NB-IoT or LTE Cat M1.

Streetlights can be used in the same manner to form an infrastructure, where, typically, Bluetooth 5 is used to extend coverage and distance between nodes in the network.

Stefan Berggren, senior product marketing manager - product strategy short-range radio, u-blox, says, "Lights (luminaires or light bulbs) in a building are ideal for creating a cloud-connected wireless infrastructure through mesh. There are many lights in a building that could serve as nodes in a mesh network. Lights are placed in a good position on the roof where these are regularly spread and often in line-of-sight of each other. Nodes in a mesh network that route the data consume more power than end nodes. Through the mesh infrastructure, you can wirelessly control lights, that is, turn these on and off and control their dimming, light temperature and colour.

"Besides connecting actual lights, a huge number of sensors, light switches and displays can be connected to the mesh network. These nodes are typically battery-operated or operated through energy harvesting, and consume very little power. Lights can also be used as beacons for positioning and advertising services."

Innovations in design technologies

Sambit Sengupta, associate director - field applications, Avnet India, says, "With the advent of smart homes, smart parking systems and smart buildings, there is a huge thrust given

to design-connected lighting. Mood lighting, health-enhancing lighting or optimum products for horticulture all require specialised designs. Graphene and violet LED technologies are contributing to newer and efficient designs and controls. Lighting with Bluetooth Low Energy (BLE) and self-healing networks is becoming commonplace."

Steve Roberts, technical director, RECOM Power GmbH, adds, "To future-proof lighting systems, designers are now incorporating multiple-format wireless controllers that can handle several different protocols. This is helped by firmware innovations such as DotDot from Zigbee Alliance that makes the application layer (bits you program to make the light do what you want) independent of radio and networking stacks (bits that handle communication).

"Also, the traditional boundary between wireless and wired control systems is disappearing with hybrid installations. For example, DALI Wireless allows designers to create wired zones that are wirelessly interlinked. Low-power AC-to-DC and DC-to-DC power supplies are suitable for powering always-on radio mesh nodes."

Mohit Kumar Chauhan, manager - pre-sales, Maksat Technologies, says, "Our solution for surveillance, Wi-Fi hotspot data networks, environmental sensor data and manageable smart LED lighting requirements is a three-in-one unit with module-based functionalities. The system can be deployed along the streets or lanes in villages, for creating a hotspot mesh network for both data access and remote surveillance. It encompasses a 360-degree pan-tilt-zoom camera for surveillance, where all functionalities are fully-managed and controlled remotely by controller